Classification of quantum graphs over complex 3×3 matrices

(Klasyfikacja grafów kwantowych nad zespolonymi macierzami 3×3)

Filip Rabiega

Master's thesis

Supervisor: Dr. Mariusz Tobolski

Uniwersytet Wrocławski
Wydział Matematyki i Informatyki
Instytut Matematyczny

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Abstract

According to all known laws of aviation, there is no way a bee should be able to fly. Its wings are too small to get its fat little body off the ground. The bee, of course, flies anyway because bees don't care what humans think is impossible.

Według wszystkich znanych praw lotnictwa, pszczoła nie powinna w ogóle latać. Jej skrzydła są zbyt małe, by unieść jej pulchne ciałko z ziemi. Pszczoła, oczywiście, lata mimo wszystko, ponieważ pszczoły nie przejmują się tym, co ludzie uważają za niemożliwe.

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Chapter 1

Introduction

1.1 Motivation

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1.3 Group theory

In the following section we will introduce some group theoretic notions that be used throughout this thesis. We assume that the Reader is familiar with the notion of a group.

Definition 1.3.1 (Dual of a group)**.** Let G be a group. The *dual group* $\hat{G}$ is the set of all homomorphisms from G into the multiplicative group of roots of unity in $\mathbb{C}$ with the group operation defined as

$$(\varphi\psi)(g) = \varphi(g)\psi(g)$$

for all $\varphi, \psi \in \hat{G}$ and $g \in G$, where the latter multiplication takes place in $\mathbb{C}$. We will often call the members of the dual group *characters.*

Remark 1.3.1. We will often call the elements of the dual group *characters.* If a group is isomorphic to its complement it will be called *self-dual.*

The following proposition will be used throughout the thesis, as almost all of the groups we will be dealing with will be finite and abelian.

Proposition 1.3.1. *If G is finite and abelian, then it is self-dual.*

Proof. First we will prove the statement for finite cyclic groups $G = \langle g \rangle$, $\operatorname{ord}(g) = n$. We will prove that the dual is cyclic of order n. Let $\chi : G \to \mathbb{C}$ be a character. Since the group is cyclic, it is entirely determined by the value it takes on its generator g. Since $g^n = e$, obviously $\chi(g)^n = 1$, so $\chi(g)^n \in \mu_n$. For each possible choice of

$\chi(g) \in \mu_n$ we get a different character. It is then easy to see that $\chi : G \to \mathbb{C}$ such that $\chi(g) = \exp(2\pi i/n)$ generates this group.

Now since every finite abelian group can be written as a finite product of finite cycles, all we need to do is check whether the statement holds for a product of two groups A, B that are themselves self-dual, i.e. $\widehat{A \times B} \cong \widehat{A} \times \widehat{B}$. This is self-evident as every character $\chi \in \widehat{A \times B}$ is determined by its restrictions to $A \times \{e_A\}$ and $\{e\} \times B$, since

$$\chi(a, b) = \chi((a, e)(e, b)) = \chi(a, e)\chi(e, b).$$

Thus $\chi \leftrightarrow (\chi|_A, \chi|_B)$ is an isomorphism. □

Definition 1.3.2 (Group action on a C^*-algebra)**.** By *group action* of a group G on a C^*-algebra A we mean a homomorphism

$$\alpha : G \to \mathrm{Aut}(A).$$

We will write $g \cdot a$ instead of $\alpha(g)(a)$ then the action in question is clear.

1.4 C^*-algebras

In the following section we will introduce some notions stemming from the theory of C^*-algebras.

Definition 1.4.1 (Linear functional)**.** A *linear functional* on a C^*-algebra A is a linear function $\psi : A \to \mathbb{C}$.

Remark 1.4.1. We will call linear functionals simply as *functionals*.

The following notions will be useful for defining an inner product on C^*-algebras. Positiveness and faithfulness are all we need to define a norm on a C^*-algebra equipped with a suitable functional.

Definition 1.4.2 (Positive functional)**.** A functional ψ on a C^*-algebra A is called *positive*, if

$$\psi(a^*a) \geq 0$$

for any $a \in A$.

Definition 1.4.3 (Faithful functional)**.** A positive functional ψ on a C^*-algebra A is called *faithful*, if

$$\psi(a^*a) = 0 \Leftrightarrow a = 0$$

for any $a \in A$.

1.4.1 Crossed-product C^*-algebra

In this section we will introduce the notion of a crossed-product C^*-algebra. We will do so in a limited manner, only to the extent to which the theory of such products is necessary for our purposes. For a more general reference we refer the Reader to Blackadar [1].

Definition 1.4.4 (C^*-dynamical system)**.** A C^*-dynamical system is a triple (A, G, α), where A is a C^*-algebra, G is a group, and $\alpha : G \to \mathrm{Aut}(A)$ is a group action on A. For a C^* dynamical system (A, G, α) with G finite we define

$$C_c(G, A, \alpha) = \{f : G \to A\},$$

i.e. the space of all functions on G with values in A.

$C_c(G, A, \alpha)$ is equipped with both a product and a $*$-operation in such a way that the action becomes "inner", i.e. $g \cdot a = u_g a u_g^*$ for all $g \in G$, $a \in A$. More precisely we define those two operations in the following way:

$$ab = \sum_{g,h \in G}$$

Definition 1.4.5 (Cross-product C^*-algebra)**.** Let A be a C^*-algebra and let G be a group acting on A. blabl

1.5 Quantum sets

Let B denote a C^*-algebra and let ψ denote a functional on this algebra. By the Gelfand-Naimark theorem, every C^*-algebra is isometrically $*$-isomorphic to a C^*-subalgebra $B(\mathcal{H})$ for some Hilbert space $\mathcal{H}$. If B is finite-dimensional, one can identify $L^2(B)$ with B and think of B as a Hilbert space with an inner product given by

$$\langle a, b \rangle_\psi = \psi(a^* b).$$

The following definition introduces one of the central notions used in this thesis. It is taken directly from Schäfer i Tobolski [3].

Definition 1.5.1 (Quantum set, Schäfer i Tobolski [3], Definition 2.1)**.** A quantum set (B, ψ) is a pair consisting of a finite-dimensional C^*-algebra B and a faithful positive functional $\psi : B \to \mathbb{C}$ such that

$$mm^* = \mathrm{Id}_B$$

holds for the multiplication map $m : B \otimes B \to B$ and its adjoint m^* given by

$$\langle m(a \otimes b), c \rangle = \langle a \otimes b, m^*(c) \rangle_{\psi \otimes \psi}$$

for all $a, b, c \in B$.

Remark 1.5.1. Recall that in general, a *comultiplication* on a C^*-algebra is a $*$-homomorphism

$$\Delta : A \to A \otimes A$$

such that the following diagram commutes:

$$\begin{tikzcd} A \arrow[r, "\Delta"] \arrow[d, "\Delta"'] & A \otimes A \arrow[d, "\Delta\otimes\mathrm{id}"] \\ A \otimes A \arrow[r, "\mathrm{id}\otimes\Delta"'] & A \otimes A \otimes A \end{tikzcd}$$

Note that in general a comultiplication is *not* unique. In our setting it is uniquely determined by the additional structure and the Riesz representation theorem for Hilbert spaces.

Remark 1.5.2. We will call any functional with the property given in the definition above a *quantum set functional* on B.

Example 1.5.1. Every finite set X gives rise to a quantum set $(C(X), \psi_X)$ with $\psi_X : C(X) \to \mathbb{C}$, $\psi_X(f) = \sum_{x\in X} f(x)$ for any $f \in C(X)$. Indeed, with this setup $m^*(e_x) = e_x \otimes e_x$ holds:

Example 1.5.2. In particular, (C^n, ψ_n) with $\psi_n(e_i) = 1$ is a quantum set. In this case...

Example 1.5.3. For any $n \in \mathbb{N}_+$, $(M_n, n\,\mathrm{Tr})$ is a quantum set, with Tr being the trace.

Recall that comultiplication

1.6 Quantum graphs

A simple graph is classically defined as a pair (V, E) with V serving as the *set of vertices* and $E \subseteq [V]^2$ serving as the *set of edges*. Such a graph could be interpreted as a matrix $A = [a_{ij}] \in \{0, 1\}^{n\times n}$ with $\{v_i, v_j\} \in E \Leftrightarrow a_{ij} = 1$; such a representation is often called the *adjacency matrix*. The key property of such matrices is that they are invariant with respect to entrywise multiplication. In turn we can use this property to generalise graphs on sets to graphs on quantum sets.

The following definitions are due to Schäfer i Tobolski [3].

Definition 1.6.1 (Quantum adjacency matrix, Schäfer i Tobolski [3], Definition 2.5)**.** Let (B, ψ) be a quantum set. A *quantum adjacency matrix* on this quantum set is a linear map $A : B \to B$ which is Schur-idempotent and *-preserving, by which we mean that it satisfies

$$m(A \otimes A)m^* = A, \qquad A(b^*) = A(b)^* \quad \text{for all } b \in B.$$

Definition 1.6.2 (Quantum graph, Schäfer i Tobolski [3], Definition 2.5)**.** A *quantum graph* on (B, ψ) is given by a quantum adjacency matrix on this quantum set.

For all intents and purposes we will identify both notions and simply call an appropriate $A : B \to B$ both a quantum adjancenty matrix and a quantum graph.

Remark 1.6.1. There are other equivalent definitions of a quantum graph, namely...

Chapter 2

Classification of quantum graphs over 3×3 matrices

2.1 Gromada's theorem

In the following section we will state...

The following theorem is due to Gromada [2].

Theorem 2.1.1 (Gromada [2], Proposition 3.1)**.** *Let A be a finite quantum set. A quantum graph...*

Bibliografia

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[2] Daniel Gromada. "Some examples of quantum graphs". W: *Letters in Mathematical Physics* 112.6 (list. 2022). ISSN: 1573-0530. DOI: `10.1007/s11005-022-01603-5`. URL: `http://dx.doi.org/10.1007/s11005-022-01603-5`.

[3] Björn Schäfer i Mariusz Tobolski. "Voltage Quantum Graphs and a Gross–Tucker Theorem for Quantum Graphs". W: (2026). arXiv: `2602.20996 [math.OA]`.